
7. When is financial education successful?

Taking stock of the new wave of field evidence

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INTRODUCTION

Financial education is a divisive topic. While such programs remain incredibly popular among policymakers, NGOs, and international organizations alike,¹ the excitement is considerably muted among academic researchers studying causal linkages.

The surge in policy interest can be traced back to the recent financial crisis and the ensuing demand for enabling better financial decision-making among individuals and households, as well as robust survey evidence on the positive association between financial literacy and prudent financial choices.² In step with this rising policy focus, the academic community jumped on the financial education bandwagon and implemented a series of field experiments to identify the causal impacts of such programs on a wide array of outcomes, ranging from savings and credit to insurance and future planning.³

The results from this first wave of field studies were astoundingly dismal. Not only did the evidence find limited average impacts of financial education,⁴ but several study managers also struggled to garner interest and demand for such programs among potential beneficiaries. In a large-scale evaluation of over 300,000 individuals in Mexico City that struggled with low take-up rates, Bruhn, Ibarra, and McKenzie (2014) find that while people did respond to monetary incentives for attendance, the marginal impacts of the program on financial outcomes were minimal. As a result, the authors assert that “people are likely making optimal choices not to attend this financial education course.” Similarly, a heavily cited early meta-analysis of financial education programs confirmed this growing consternation among academics, with Fernandes, Lynch, and Netemeyer (2014) concluding that “financial education interventions studied explained only about 0.1% of the variance in the financial behaviors studied, with even weaker average effects of interventions directed at low-income rather than general population samples.”

These early results deflated the outlook and value of financial education in the academic community, with a general acceptance that such programs simply do not work. Unfortunately, despite fundamental issues with the sampling and estimation framework of Fernandes, Lynch, and Netemeyer (2014),⁵ this negative view has prevailed over the last few years with only lukewarm interest in financial education as an effective reform or even as a topic of further research.

Nevertheless, many researchers have continued to work on financial education evaluations and recent evidence on its impact suggests that such broad skepticism is unwarranted. In a new wide-ranging meta-analysis of 76 RCTs on financial education, Kaiser et al. (2020) find that

[financial education] treatment effects are economically meaningful in size, similar to those realized by educational interventions in other domains, and are at least three times as large as the average effect documented in earlier work. These results are robust to the method used, restricting the

sample to papers published in top economics journals, including only studies with adequate power, and accounting for publication selection bias in the literature.

Hence, not only are newer studies revealing statistically significant impacts of financial education, but the effect sizes detected are also large. These findings paint a substantially different and much more promising outlook for financial education than previously accepted.

While Kaiser et al. (2020) offer valuable insight on the common effects of financial education that can be derived from the recent literature, the focus of this review is to unpack where these impacts are coming from. Specifically, this review seeks to highlight particular features, target groups, timing choices, and complementary interventions that can make financial education programs successful.

Despite the reported successes due to a maturing literature, the analysis in this review underlines important outstanding tensions across studies that show potentially contradictory results on some financial outcomes such as credit decisions. As such, the review does not herald financial education as a panacea, rather it poses important open questions for future research that can help both policymakers and academics make informed choices going forward.

WHAT PROBLEM DOES FINANCIAL EDUCATION SEEK TO SOLVE?

The fundamental underlying question that motivates financial education research is this: why do people make sub-optimal financial decisions in their lives? This is a question of pressing policy and academic relevance, given the rapid proliferation of new financial products and services around the world.

Various strands of research in the financial inclusion space have shown that there may be several, interlocking reasons. First, the structure of the financial system is difficult to navigate for the modal user, let alone for someone with low financial literacy. Financial products are complex and difficult to evaluate. When taking out a loan, for example, a borrower must calculate interest rates and compare different types of credit. Anticipating liquidity needs is likewise difficult, particularly if income is volatile, as it is for many segments of the developing world population. Second, individuals are simply not very good at inter-temporal decision-making; studies show that people tend to lack self-control (Ashraf, Karlan, and Yin, 2006), procrastinate about financial decisions (World Bank, 2017), and are deterred by small and inconsequential “channel” factors (Bertrand, Mullainathan, and Shafir, 2006). Finally, there are significant informational asymmetries on the supply side, where research has shown that financial advisors lack disclosure and transparency with their clients (Giné, Cuellar, and Mazer, 2017).

Given these demand and supply barriers, the concept of financial education is appealing. Not only can a more financially savvy individual better discern financial products and services, but financial education may also help overcome behavioral constraints if the costs of adoption of financial products are lowered. Furthermore, financial education can help individuals navigate the often-opaque financial market landscape more effectively. With the rapid proliferation of new financial products and services owing to technological advancements, the need for informed financial decision-making is increasingly pertinent.

Policymakers in both developed and developing countries have latched on to this promise of financial education with incredible enthusiasm. Comprehensive national-level initiatives and

programs funded by the World Bank, DFID, OECD, and other development donors have fast-emerged all over the world in the last decade. This enthusiasm is supported by ample research evidence that indicates a strong correlation between good financial decisions and levels of financial literacy. Studies in developed countries show that individuals with lower financial literacy tend not to plan for retirement (Lusardi and Mitchell, 2007a), borrow at higher interest rates (Lusardi and Tufano, 2008; Stango and Zinman, 2006), acquire fewer assets (Lusardi and Mitchell, 2007b), and participate less in the formal financial system (Alessie, van Rooij, and Lusardi, 2011; Hogarth and O'Donnell, 1999). Similar correlations have been borne out in developing country studies as well (e.g. Cole, Sampson, and Zia, 2011; Klapper and Lusardi, 2012). This evidence suggests a pathway whereby financial education boosts financial knowledge, which in turn enables better financial decisions.

Moving from correlative statements to causal ones, however, is difficult. Making survey-based inferences on financial decision-making suffers from issues of endogeneity, omitted variable bias, and directionality. For instance, is it that high financial literacy leads to better financial decisions, or is it that better decision-makers are instinctually better at making financial decisions? This identification challenge is echoed in observational studies of the effects of financial education, which tend to have mixed results. Bernheim and Garrett (2003) use household survey data to analyze the effect of employer-based financial education on the savings behavior of employees, both in general and for retirement. Their analysis finds a positive correlation between education programs and savings, with sufficient controls to suggest that the relationship is causal. Meanwhile, Cole and Shastry (2009) replicate and extend a Bernheim, Garret, and Maki (2001) study on financial education mandates in US high schools using a much larger sample from census data. Their results contradict the original finding that high school education impacts future savings and investment.

Shifting to experimental studies also comes with a unique set of challenges, as the early wave of field experiments did not find significant causal effects of financial education. While the focus of this review is on the new wave of evidence, it is constructive to briefly summarize earlier studies and highlight likely culprits for the lack of impact.

EARLY WAVE (LACK OF) EVIDENCE

A common theme among early field experiments on financial education was that they evaluated primarily classroom-based programs that relied on standardized curricula with some local contextualization. For example, in the first field experiment on financial education in developing countries, Cole, Sampson, and Zia (2011) adapted a global curriculum developed by a consortium of Microfinance Opportunities, Citigroup Foundation, and Freedom from Hunger to the Indonesian classroom context. On the heels of a growing wave of financial education enthusiasm, this study found a precisely estimated *zero* average effect of financial education on savings decisions. A follow-up study in India that employed professionally produced and locally developed video content in classrooms also found that instruction alone did not lead to a meaningful impact on financial behavior (Carpena et al., 2019). Likewise, an ambitious experiment in Mexico (Bruhn, Ibarra, and McKenzie, 2014) sought to study the impact of adult financial education workshops; however, despite offering various incentives for attendance, interest in the workshop was menial. Moreover, even in cases where early studies did find modest effects of classroom-based financial education, the effects tended to

dissipate over time (Fernandes, Lynch, and Netemeyer, 2014) and showed substantial variation across study contexts (Miller et al., 2015).

This evidence highlights the pitfalls of standard, classroom-based instruction for adults and its inability to attract participants and engage them in a meaningful way. In hindsight, the scope and reach of even the most well-designed classroom financial education programs are likely limited. On the supply side, these programs are difficult to design and expensive, and finding skilled educators is a challenge. On the demand side, it is difficult to attract participants and retain interest throughout the duration of the course. These challenges are only magnified by human behavioral patterns: inattentiveness, lack of attention toward non-salient content, procrastination, and various other channel factors – all potential impediments to moving from financial knowledge to behavior change.

A fundamental issue that financial education programs butt against is that financial behaviors are very difficult to change, especially among adults, which can stymie good intentions of the program. In addition, even if participants do come away from financial education training understanding the value of financial decisions (e.g. opening a bank account and saving is a good idea), there may be practical factors that inhibit them from doing so (e.g. not having disposable income). Moreover, the binding constraint may not be financial education but rather the income needed to save in the first place. In such circumstances, financial education can be most effective when the lessons target outcomes with lower adoption hurdles or when complementary limiters are removed. For example, research has shown that simple text message reminders are an effective way to induce timely loan repayment and achieve savings goals (Karlan et al., 2016) because they target the behavioral trait of human inattentiveness. Similarly, displaying information about how much the borrower will have to pay in monetary terms as opposed to interest rate terms improves borrowing decisions and reduces the use of high-interest loans (Bertrand and Morse, 2011; Hastings and Tejada-Ashton, 2008). In recent years, researchers have begun to advance similarly innovative programs in the financial education space that attempt to overcome many of the delivery and complementary constraints that thwarted impacts in earlier studies.

A final point about early studies is that even though *average* treatment effects were muted, there were sub-groups that did benefit from financial education. For example, while Cole, Sampson, and Zia (2011) find no average effects, the results do show that illiterate individuals and those with low initial levels of financial literacy did benefit substantially from the financial education program. Specifically, while illiterate individuals were only a tenth of the total study sample – which explains the muted average effects – the heterogeneous results show a 12.3 percentage point improvement in the likelihood of opening a bank account over a base of 5 percent in the control group. This is a 246 percent improvement over the control group, an effect size that is difficult to ignore.

The point about heterogeneity is an important one. From a policy perspective, if financial education enables even a small proportion of the target population to take up savings or make responsible money decisions, then labeling the policy an outright failure is misguided. This is especially true if the heterogeneity is among high-risk groups or vulnerable populations where a little money saved or avoidance of bankruptcy may be life-altering. What this does imply, however, is a need for better targeting of financial education, a theme that is picked up in the new wave of research.

FINANCIAL EDUCATION SUCCESS IN THE NEW WAVE OF RESEARCH

The new wave of research on financial education takes seriously issues of measurement, targeting, timing, and combining financial education with simple add-ons that may help alleviate complementary constraints. At the same time, many key outstanding questions remain. This section highlights several studies which have adopted these new approaches and the lessons learned.

Measurement

Perhaps the most fundamental question when evaluating financial education programs is how to measure financial literacy itself. Indeed, this is not an easy choice as it is unclear what exactly comprises the term “financial literacy.” It can encompass concepts ranging from financial awareness and knowledge, including knowledge of financial products, institutions, and concepts; financial skills, such as the ability to calculate compound interest payments; and financial capability more generally, in terms of money management and financial planning. In practice, however, these notions frequently overlap.

Measurement criteria may also vary depending on the income level of the country. In high-income countries, for instance, financial education is often viewed as a complement to consumer protection. One of the primary goals of financial education is therefore to equip individuals with the capability to navigate a complex array of financial products, including pensions and mortgages, and to make sound financial decisions. The importance of doing so has been further underscored in recent years by the financial crisis and the continued shifting of retirement planning responsibilities from the public sector to individuals.

In low-income countries, however, financial outreach is much more limited, and more sophisticated consumer products are typically accessible only to a small percentage of the population. The role of financial education in increasing access to and take-up of financial services therefore receives more focus and offers a potentially valuable policy tool to stimulate financial inclusion in these countries.

Given the differences in the objectives of financial education across settings, it naturally follows that measurement metrics likely also need to be customized to the local context. Many financial literacy surveys in high-income countries exhibit two approaches to measuring financial literacy (OECD, 2005). The first measures respondents’ understanding of financial terms and their ability to apply financial concepts to particular situations, and the second asks respondents for a self-assessment of their financial understanding and knowledge as well as their perceptions and attitudes toward financial instruments and decisions. The survey questions that are the standard measure of financial literacy, however, are those developed by Lusardi and Mitchell (2008) on numeracy, inflation, and diversification.⁶ These questions have been included in various surveys of financial literacy in many parts of the world, including the US, the Netherlands, Italy, Germany, and New Zealand. They have likewise been adapted to measure financial literacy in research conducted in developing countries, such as Indonesia (Cole, Sampson, and Zia, 2011), South Africa (Cole, Shapiro, and Shastry, 2010), Sri Lanka (de Mel, McKenzie, and Woodruff, 2009), and Mexico (Hastings and Tejada-Ashton, 2008).

While Lusardi and Mitchell (2008) provide a standard set of questions to assess financial literacy, these measures are not necessarily comprehensive and may not be appropriate in the

developing country context. Indeed, Lusardi and Mitchell (2008) state that it is “imperative to expand the range of measures of financial literacy, so as to better evaluate the types of problems that people find most difficult.” Developing countries are typically characterized by high poverty, low access to finance, and a lack of consumer protection measures. Hence, the environment is quite different than the developed world and the concept and definition of financial literacy ought to also be different. For instance, in an environment where most households are uneducated and hold informal savings, it may be important to assess financial literacy based on knowledge of bank account opening requirements, as opposed to the ability to calculate interest rates.

This is the thesis of recent work by Carpena and Zia (2020) that proposes a new approach to measuring financial literacy in the developing country context. They develop and test a three-dimensional measure of financial literacy that assesses numeracy skills, basic financial awareness, and attitudes toward financial decisions. Numeracy skills are those that involve calculating interest rates, adding income, and similar computations; basic financial awareness consists of knowledge about the details of a budget, savings account, and loan processing fees; attitudes toward financial decisions involve individual perspectives about the benefits of financial products.

This broader measure of financial literacy is more in line with the financial realities of individuals in the developing world. While many uneducated individuals may not be able to respond correctly to computationally heavy survey questions, they may still be aware of the financial choices available to them, may know where to turn to for financial help, and may be able to offer intelligent financial product advice to their friends and families. The Carpena and Zia (2020) approach captures these broader channels of financial knowledge, and such a composite measure is now becoming more common in financial access studies in developing countries.

The additional benefit of a broader measure of financial literacy is the ability to identify the impacts of financial education programs on different dimensions of financial knowledge. In this vein, Carpena and Zia (2020) apply formal causal mediation analysis in the spirit of Imai et al. (2011) and Acharya et al. (2016) to identify the relative merits of numeracy, awareness, and attitudes as causal mediators of financial outcomes in a field experiment setting in India.⁷ Their findings are stark – numeracy does not mediate any of the identified impacts of financial education on any of the financial outcomes measured. By contrast, awareness and attitudes are significant channels for improvements in simple financial decisions such as budgeting, while only attitudes play a prominent role in improving more complex financial actions such as opening a formal savings account. Taken together, these findings suggest that relying on questions related to numeracy is unlikely to connect the primary link in the causal chain of financial education impact.

Other studies discussed below have adopted the same or related broader approaches to measuring impacts on financial literacy. Clearly, the measurement question is still an open one, with room for improvement in how researchers measure financial literacy. A portfolio or repository of questions that researchers can use in a pilot phase to assess contextual viability prior to study implementation is a promising steady state to strive for. More research is needed to populate this repository further with a list of relevant questions and dimensions of financial knowledge.

Targeting Youth

One of the most promising avenues of new research on financial education is among school-aged youth. The focus on youth and especially students is important and policy-relevant: first,

students in school are just beginning their interactions with the financial sector, and good financial habits formed at this early age are likely to reap benefits throughout adulthood. Second, the potential for positive impact among this segment is high for the simple reason that teaching financial concepts to a school-going population is likely easier and more rewarding than teaching adults. Children attend school every day to learn and are primed to absorb, recall, and apply what they are taught. Adults on the other hand are more difficult to teach and may have years of bad habits and sticky preferences that are averse to change. Finally, there may be potential knowledge spillovers, in which students modify not only their own behaviors but also influence their household's financial decisions.

Several carefully executed RCTs have emerged over the last few years that focus on identifying the impact of financial education on high school students. Bruhn et al. (2016) conducts a large-scale randomized evaluation of a school-based financial education program in Brazil with 892 schools in six states and nearly 25,000 students. The program involved weaving financial education topics into the regular school curriculum over an evaluation period of three full academic semesters and 18 months of instruction. The findings show significant improvements in financial knowledge and attitudes, improvements in savings and budgeting behavior, higher scores on measures of financial autonomy and intentions to save, lower likelihood of school dropout, as well as positive spillovers on parents. A key finding on distributional impacts is that the program helped students of all abilities move up – specifically, “the proportion of students that performed exceptionally well increased by 28 percent and the proportion of students that performed exceptionally poorly decreased by 26 percent.” Corroborating evidence comes from Peru, where Frischno (2020) evaluates a similar program with a sample of 300 high school students. This study uses both survey and credit bureau administrative data to find significant reductions in credit delinquencies for students and teachers up to two years later. The administrative match is imperfect, however, and the evaluation period is short, but these are promising short-term findings on spillover effects for instructors.

A key focus among other financial education studies for high school students is on inter-temporal choices. Lührmann, Serra-Garcia, and Winter (2018) evaluate a 90-minute training module on shopping planning and savings among German high school students and measure students' inter-temporal choices using an incentivized experiment. The study finds that participation in the program improves time-consistent choices related to payments, but it does not measure the program's effect on financial knowledge or real behavior. Similarly, Sutter et al. (2020) study a financial education program among German high school students and finds the program led to higher patience and lower present bias, which the authors argue helps explain an important channel of financial education impact. More direct evidence on inter-temporal choices comes from Alan and Ertac (2018) that evaluates a school curriculum focused on future selves and forward-looking behavior in Turkey. Like the other studies, the findings show higher patience among students in inter-temporal decision tasks, both in the lab and in surveys.

Not all evidence on the impact of financial education in schools is positive, however. Berry, Karlan, and Pradhan (2018) study a financial education program for primary and junior high students in 135 Ghanaian schools over a ten-month period. The results do not show significant improvements in financial knowledge, aggregate savings, or social and psychological measures. It is unclear, though, how exposed students are to financial decision-making at such a young age, and eliciting reliable and consistent responses in this age group can be challenging.

Further cautionary evidence on financial education in high schools comes from the first study in Brazil, i.e. Bruhn et al. (2016). Alongside the many positive effects highlighted earlier, the study also finds a higher likelihood among students of using expensive credit to make consumer purchases.⁸ The authors hypothesize that this may be due to a multi-tasking problem and willpower depletion where the curriculum urged students to improve financial choices on all dimensions – savings, budgeting, spending – and that the simultaneous emphasis on these fronts may have been overbearing. This evidence on likely poor credit decisions contradicts the findings in other studies on better inter-temporal planning. More longer-term research is needed to resolve this tension in the literature, and the authors of the Brazil study are currently pursuing a ten-plus-year follow-up to understand the long-term impacts of such student choices.

Timing and Teachable Moments

“Teachable moments” is a mantra repeated across the policy world when it comes to financial education (e.g. Center for Financial Inclusion, 2016), and one that is now gaining traction and support from rigorous academic research. Indeed, individuals are more likely to be more responsive to new educational material related to their finances at precisely the times when it is most valuable, i.e. a teachable moment. This insight has strong implications for how financial education programs should be designed, in particular, shifting the focus away from generic instruction to *timely* instruction along the value chain of beneficiary outcomes.

One such teachable moment is at the time of migration, when the migrant worker is likely to receive a positive income shock. Doi, McKenzie, and Zia (2014) conducted a field experiment among Indonesian migrant workers and their families with three randomly selected treatment groups: one group offered a financial education curriculum focused on remittances, savings, and money management to the migrant workers; another group offered the same course to the families of the migrant workers; and the third group offered the course to both the migrant workers and their families. The results find that the training led to more financial planning and budgeting, and a 26 percent improvement in the likelihood of savings compared to 40 percent in the control group when it was offered to both the migrant worker and the family. The number of remittances saved also doubled in this group compared to the control. In contrast, the results do not find significant improvements in outcomes when the program was offered just to the worker.

This latter result is echoed in another work by Gibson, McKenzie, and Zia (2014) that studies the impact of teaching migrants in New Zealand and Australia about remittance costs and finds no change in the frequency or level of remittances. Barcellos et al. (2016) show similar muted impacts among migrants in the US. Seshan and Yang (2014) also find similar null average effects among migrant Indian workers in Qatar; however, the analysis detects significant increases in savings and remittances as well as joint family decision-making among those with low pre-treatment savings levels.

One additional study on financial education for migrant workers is Abarar, Barua, and Yang (2020) which investigates complementarities between financial education and financial access among transnational households in the Philippines. While the study does not find significant complementarities, the results do show improvements in borrowing behavior and substitution toward formal borrowing as a result of financial education. Nevertheless, the study cannot disentangle whether this effect is due to advice given on financial planning, facilitation of goal setting, or simple information provision about borrowing choices.

Overall, these studies highlight that financial education can trigger a teachable moment for migrant workers but only when it is provided to both the sender and receiver of remittances or when it is carefully targeted at those that have low savings to begin with. Therefore, not only is timing important in the realm of teachable moments, but *who* is targeted is equally critical. These nuances are important for the success of financial education going forward, and more research is needed to understand other mechanisms underlying its success. Furthermore, greater research is needed on the durability of instruction at teachable moments – i.e. do the knowledge and behavior gains persist, or do they fade along with the novelty and newness of a teachable moment?

Complementary Interventions

Related to the last point about mechanisms, recent research on financial education has focused on understanding whether simple and inexpensive add-ons to financial education can lead to success and a viable policy proposition.

Carpena et al. (2019) pose precisely this question with a field experiment in India where the study design adds simple financial goal setting and short counseling sessions to a financial education course in separate and combined study arms. The findings confirm earlier null effects of financial education classes alone, however, they also show that adding goal setting and financial counseling has significant and consequential effects on financial behavior related to savings, budgeting, and borrowing. For instance, adding low-cost counseling to financial education led to a 45 percent increase in the likelihood of a formal savings account compared to 29 percent in the control group. Theodos, Stacy, and Daniels (2018) similarly find positive and significant impacts of one-on-one financial coaching on money management, debt, savings, and perceptions of financial well-being. Finally, Steinart et al. (2018) present evidence of the value of financial education when complemented with a psychological parenting intervention in South Africa. The results show a 65 percent improvement in the propensity to save over the control group average of 23 percent, as well as lower borrowing rates and reductions in financial distress. Thus, complementing financial education with related programs can provide the impetus for success.

Another way to complement financial education is through gamification and experiential learning, which allows for more engagement through practice and interaction, as well as through learning by doing. As Attanasio et al. (2019) point out, gamification of financial education content may be particularly useful in low-literacy contexts, which enables greater content understanding and retention. The authors evaluate a tablet-based financial education program with gamified presentation on financial outcomes of mostly female recipients in Colombia and find significant positive impacts on knowledge, attitudes, practices, and financial health over two years later – compared to the control group propensity of 43 percent, the treated group were 26 percent more likely to have formal or informal savings by the third follow-up survey. Similarly, Calderone et al. (2018) study the impact of a financial education course in India that was augmented with comics and audio-visual tools and find a 49 percent improvement in saving amounts among branchless bank customers. Finally, Abel, Cole, and Zia (2021) evaluate an experiential learning game where messages about gambling avoidance underpin a carefully designed dice game that is offered to rural South African households. The results show significant reductions in lottery ticket purchases a year later with just a five-minute interactive game.

Overall, these emerging findings show that financial education can be made successful with light repackaging, either through complementary add-ons or by changing the instruction

interface. More research is needed to understand the full range of complementary interventions. On gamification and experiential learning, more research is needed on how to scale these interventions so they can reach a much wider audience, especially in developing countries, either through unassisted technology or other media.

Delivery Channels and Scale Up

The ultimate policy success of development interventions can be tied to their ease of delivery and viability for scaling up to a larger audience. One of the key determinants of the failure of early financial education impact studies was precisely that they were difficult to deliver. After all, finding skilled and engaging instructors is not easy, delivery at the local scale is expensive and time-consuming, and developing and localizing content incurs consultancy fees and expensive pilots. In addition, the demand side is also challenging with standard classroom sessions resulting in low engagement, disinterested audiences, and drop-offs in attendance.

Research on financial education that targets both the supply- and demand-side barriers has now started to emerge in the literature. When a lack of salient content drives a lack of interest in financial education, educational entertainment, i.e. “edutainment,” may be the answer. Nearly every household in the world has access to television either in their own homes or in their communities. As such, edutainment casts a much broader net than any other delivery media. Furthermore, edutainment has the advantage of a captive audience, hence the demand-side constraints are also relaxed. While there is considerable evidence on the effectiveness of television in influencing behaviors related to health and domestic violence issues (see, for example, Jensen and Oster, 2008), Berg and Zia (2017) fill an important gap in our understanding of financial education delivered through mass media. Working with a nationally televised show that incorporated messages on debt management over a two-month-long storyline, the authors find significant improvements in borrowing and gambling behavior, two aspects of financial decision-making that were emphasized in the show. The reported effect sizes are large – conditional on borrowing the treated group is 68 percent more likely than the control group to borrow predominantly from a formal bank, 23 percent more likely to use expensive shop credit, and 17 percent less likely to spend money on gambling.

Another way to address the supply and demand constraints in financial education is to decentralize teaching responsibility. Hakizimfura, Randall, and Zia (2020) present evidence from a field experiment in Rwanda where members of 200 Savings and Credit Cooperative Associations (SACCOs) were provided comprehensive teacher training and then asked to deliver the content to their respective community constituents. The findings show stark differences based on who within the SACCO community is provided the teacher training. Specifically, the experiment design called for one group where SACCO loan officers were required to act as teachers and another group where SACCOs had discretion on who to send for the teacher training. The study finds significant effects on financial understanding, planning, and savings deposits, but only among the latter group.

These results suggest that local discretion in the selection of teachers is an important key to financial education success. However, identifying the precise types of incentives and levers for local teacher engagement is critical for the success of decentralized delivery and hence an important and open question for future research in financial education. Likewise, separating the mechanism of impact of financial education through mass media in an experimental setting is a compelling topic for future research.

Gender and High-Risk Groups

A few studies in the growing financial education literature focus on gender and high-risk recipients of training. Pellegrina et al. (2019) study the impact of a financial education program aimed at female borrowers at a non-profit microfinance institution in India and find a significant increase in savings and improved punctuality of loan payments. Bhutoria and Vignoles (2018) study a similar program focused on rural women in India and find modest improvements in savings. Koomson, Villano, and Hadley (2020) provide a gendered perspective on a financial education course in Ghana, where they find a 34 percent increase in female account ownership compared to one in four women in the control group who have an account – a sizeable improvement. However, they also find that these extensive margin changes do not translate into higher savings for women, with only male beneficiaries showing improvements on the intensive margin. These results suggest a need for more focused financial education to close the gender gap in financial account ownership and usage.

Hetling, Postmus, and Kaltz (2016) provide evidence on a financial education program targeted at survivors of intimate partner violence (IPV) in the US and Puerto Rico through a field experiment. The findings show strong effects on financial knowledge and behaviors, which indicates support for expanding such programs to other settings. Similarly, Haynes, Haynes, and Weinert (2011) study a financial education program for chronically ill rural women in the US that was delivered online. The findings show that exposure to online lessons on finance and access to discussion groups with experts and social peers led to an almost 30 percent increase in financial knowledge scores, though financial behaviors were not measured.

Overall, much more research is needed to understand how financial education can be particularly helpful for women and high-risk social groups, especially in closing gaps in financial inclusion and participation. This line of inquiry circles back to the earlier point about heterogeneity – measuring the success of financial education by *average* effect sizes may be the wrong yardstick, especially when such programs can lead to critical welfare improvements for a subset of the population. Greater research is needed to identify such heterogeneous beneficiaries so financial education can be better targeted and timed.

CONCLUSION

Recent research illuminates that financial education programs can be an effective tool in improving financial decision-making when delivered at the right time, to the right audience, through the right channels, and in combination with other interventions to solve multiple constraints that individuals face. But while this recent work is promising, the evidence on financial education success is still nascent with many important outstanding questions that are ripe for future research.

While this chapter provides insight into several of these future research topics inline, two important areas are worth emphasizing in the conclusion. First, like many other areas of field research, we know very little about the long-term impacts of financial education. Do the short-term impacts identified in the new studies last? If they dissipate, can they be reactivated through simple channels? Over time, what are the multiplicative benefits of financial education on welfare more broadly?

Second, technological advancements and new data use and repurposing are opening new avenues for outreach and communication to a multitude of target groups that were previously

unimaginable (World Bank, 2021). Digital platforms and machine learning approaches are starting to offer cutting-edge and innovative channels for engaging delivery of education content. These methods promise not only inexpensive scale-up but also personalized content. Much more research is needed on evaluating the success and viability of these channels as vehicles of financial education.

NOTES

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1. The United States adopted a President's Advisory Council on Financial Literacy in 2008 to help promote financial education at all levels of the economy; the UK government mandated compulsory financial education in schools from 2012 onward; the Indonesian government declared 2008 as the year of financial education; the Reserve Bank of India launched a series of financial literacy and counseling centers across the country in 2007; Brazil and many other developing countries have incorporated national strategies for improving financial education. These are only a few examples.
 2. For a summary of these findings, see, for example: Xu and Zia (2012), Lusardi and Mitchell (2014), World Bank (2017).
 3. For literature reviews, see, for example: Fernandes, Lynch, and Netemeyer (2014), Hastings et al. (2013), Miller et al. (2015), Kaiser and Menkhoff (2017), and most recently Kaiser et al. (2020).
 4. See, for example: Cole, Sampson, and Zia (2011), Chong et al. (2010).
 5. The Fernandes, Lynch, and Netemeyer (2014) meta-analysis included only 13 early RCTs on financial education and the analysis did not account for substantial programmatic differences across settings. The "variance explained" metric also underestimates the economic significance and size of reported effects. Subsequent meta analyses such as Miller et al. (2015) and Kaiser and Menkhoff (2017) apply more nuanced approaches to sample selection and estimation, with more positive findings on the impact of financial education. See Kaiser et al. (2020) for a detailed discussion and comparison of estimation strategies across these earlier studies.
 6. The financial literacy questions developed by Lusardi and Mitchell (2008) are the following: (1) Suppose you had \$100 in a savings account and the interest rate was 2 percent per year. After five years, how much do you think you would have in the account if you left the money to grow: more than \$102, exactly \$102, or less than \$102? (2) Imagine that the interest rate on your savings account was 1 percent per year and inflation was 2 percent per year. After one year, would you be able to buy more than, exactly the same as, or less than today with the money in this account? (3) Do you think that the following statement is true or false? "Buying a single company stock usually provides a safer return than a stock mutual fund."
 7. Causal mediation analysis relies on a strong identifying assumption that the mediators are statistically independent of the potential outcome. See Carpena and Zia (2020) for a detailed discussion, application of various identification methods, as well as robustness with sensitivity analysis. With multiple mediators, mediation analysis can be helpful when comparing the *relative* magnitudes of the different mediation channels. This comparison relies on a slightly weaker identification assumption that there is no *differential* confounding effect across the different mediators. Moreover, if we consider motivation or ability as an unobserved confounder, this assumption states that the confounder affects financial awareness the same way it affects numeracy or attitudes. As such, this comparison provides suggestive evidence on which mediating channels are important.
 8. Kaiser and Menkhoff (2020) report similar findings of higher savings but also higher likelihood of loan delinquency among small-scale retailers in Uganda who were exposed to a financial education program with an active learning component.

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